Exploratory Data Analysis & Basic Model

📊 Data Understanding

The dataset contains information about schools, including:

- Student performance (% Math, Science, Language scores) .
- Teacher-related factors (experience, training, teacher-student ratio).
- Infrastructure facilities (electricity, internet, toilets, library).
- Socio-economic indicators (parent literacy rate, marginalized students).
- The main target variable created was:
  Average_Score (average of all subject scores).

🧹 Data Cleaning

- Missing values were handled using **mean imputation.**
- Duplicate records were removed to ensure data quality.
- Data was prepared for analysis and modelling.

🔍 Key Insights

- Students performing well in Math also perform well in Science.
- Urban schools generally perform better than rural schools.
- Higher teacher-student ratio negatively affects performance.
- Better infrastructure improves student outcomes .
- Parent literacy rate has a positive impact on student performance.

🤖 Basic Machine Learning Model

A **Linear Regression model** was built to predict Average Score using:

- Teacher-related factors
- Infrastructure availability
- Socio-economic indicators

The model was evaluated using:

- **MAE (Mean Absolute Error)** → prediction error
- **R^2 Score** → model accuracy.

🎯 Conclusion

The analysis shows that student performance is influenced by a combination of school infrastructure, teacher quality, and socio-economic conditions. EDA helped identify important patterns, and the basic model demonstrated that these factors can reasonably predict student performance.